

Task 7

As part of your cloud training, you are assigned a **mini project** called the “**Server Mood Game**”.

This assignment is designed to help you understand how **EC2, Application Load Balancers, and AWS Lambda** work together in a simple and interactive way.

Project Concept: Server Mood Game

When a user accesses the application using the **Load Balancer DNS**, the response should display a **server mood** based on which EC2 instance handled the request.

To make the application more dynamic and interesting, a **Lambda function** will also be invoked to add an additional message to the response.

Each page refresh should show a **different combination** of:

- EC2 server mood
- Lambda-generated message

What the User Should See

On refreshing the page or clicking a button, the output should look similar to the following:

Example 1

😊 I am happy and healthy!
Handled by: Web-Server-1

Lambda Message:
Lambda says: Keep calm and scale on!

Example 2

😓 I am tired but still serving traffic!
Handled by: Web-Server-2

Lambda Message:

Lambda says: High traffic detected!

Each refresh may show a different result because:

- Traffic is routed via a Load Balancer
- Lambda returns a dynamic response

What You Need to Build

EC2 & Load Balancer

- Create **two EC2 instances** (preferably in private subnets)
- Install **Apache / httpd** on both instances
- Configure each EC2 instance to return a **different mood message**
- Create an **Application Load Balancer (ALB)**
- Attach both EC2 instances to the ALB target group
- Ensure the application is accessible **only via the ALB DNS**
- EC2 instances should **not be directly accessible via public IP**

Lambda Integration (Mandatory)

- Create a **Lambda function** that returns a simple dynamic response, such as:
 - A random emoji
 - A random cloud-related message
 - A timestamp or random number
- The EC2 application should **invoke the Lambda function**
- Combine the **EC2 mood message** and **Lambda response** and display them on the web page

How the Game Works

1. User opens the application using the **ALB DNS**
2. ALB routes the request to one EC2 instance
3. EC2 processes the request and calls the Lambda function
4. Lambda returns a dynamic message
5. The combined response is displayed to the user
6. On multiple refreshes, different outputs should appear

Output:

The task will be considered complete if:

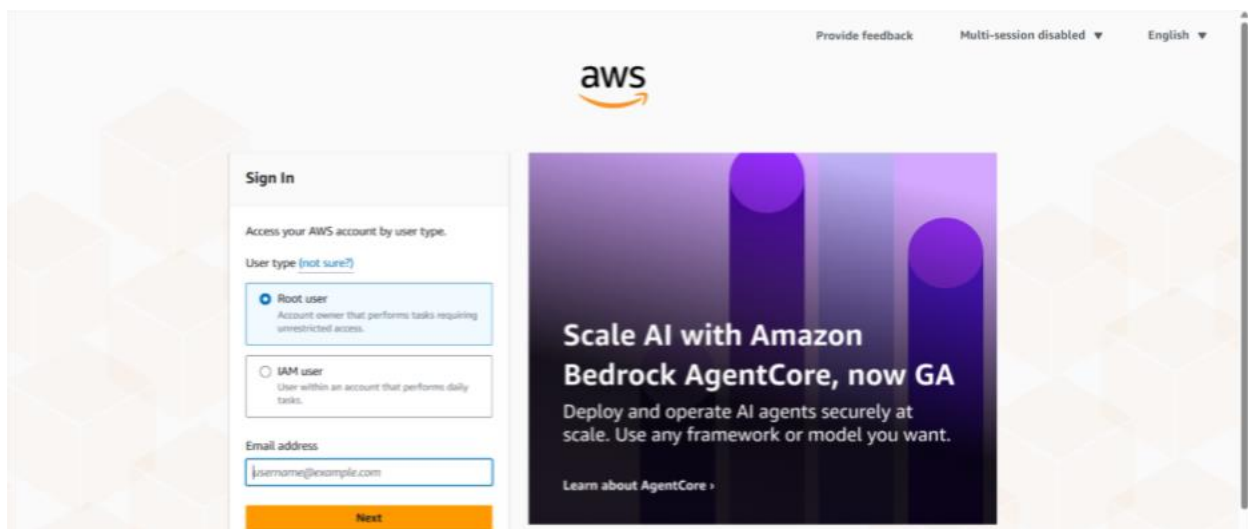
- Application is accessible using the **Load Balancer DNS**
- Each EC2 instance displays a **unique mood**
- Lambda function is successfully invoked
- Lambda output is visible in the response
- Multiple refreshes show **different combinations**
- EC2 instances are securely deployed and not publicly exposed

Solution:

Login to your AWS account with credentials and required verification.

1. (<https://signin.aws.amazon.com/>)

My account – AWS Management Console- Sign in



2. Create the Network (VPC & Subnets)

2.1 Create a VPC

- Go to VPC → Create VPC
- Name: server-mood-vpc
- IPv4 CIDR: 10.0.0.0/16
- Create

2.2 Create 4 subnets:

Public Subnets (for ALB)

- Public-Subnet-1: 10.0.1.0/24 (AZ-a)
- Public-Subnet-2: 10.0.2.0/24 (AZ-b)

Private Subnets (for EC2)

- Private-Subnet-1: 10.0.3.0/24 (AZ-a)
- Private-Subnet-2: 10.0.4.0/24 (AZ-b)

2.3 Create Internet Gateway (*servermood-igw*)

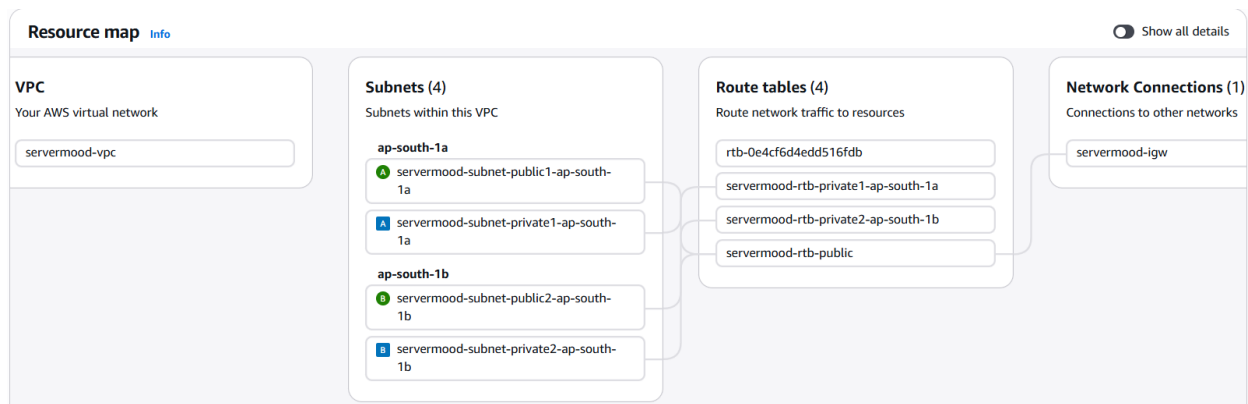
2.4 Route Tables

Public Route Table

- Route: 0.0.0.0/0 → Internet Gateway
- Associate public subnets

Private Route Table

- No internet route
- Associate private subnets



3. Create IAM Role for EC2 to Call Lambda

Create IAM Role

- IAM → Roles → Create Role
- Trusted entity: **EC2**
- Permissions:

- *AWSLambdaBasicExecutionRole*
- Add inline policy

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.

Use case
☒ **EC2**
Allows EC2 instances to call AWS services on your behalf.

Permissions policies (1/1119) [Info](#)

Choose one or more policies to attach to your new role.

Filter by Type: All types (4 matches)

Policy name	Type	Description
☒ AWSLambdaBasicExecutionRole	AWS managed	Provides write permissions to CloudW...
☐ AWSLambdaBasicExecutionRole-4c80ee86-a020-475...	Customer managed	-
☐ AWSLambdaBasicExecutionRole-613ead41-1adc-4be...	Customer managed	-
☐ AWSLambdaBasicExecutionRole-692a249f-d87b-430...	Customer managed	-

► **Set permissions boundary - optional**

[Cancel](#) [Previous](#) [Next](#)

Specify permissions [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

Visual **JSON** Actions

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "lambda:InvokeFunction",
7       "Resource": "*"
8     }
9   ]
10 }
11

```

Edit statement

Select a statement

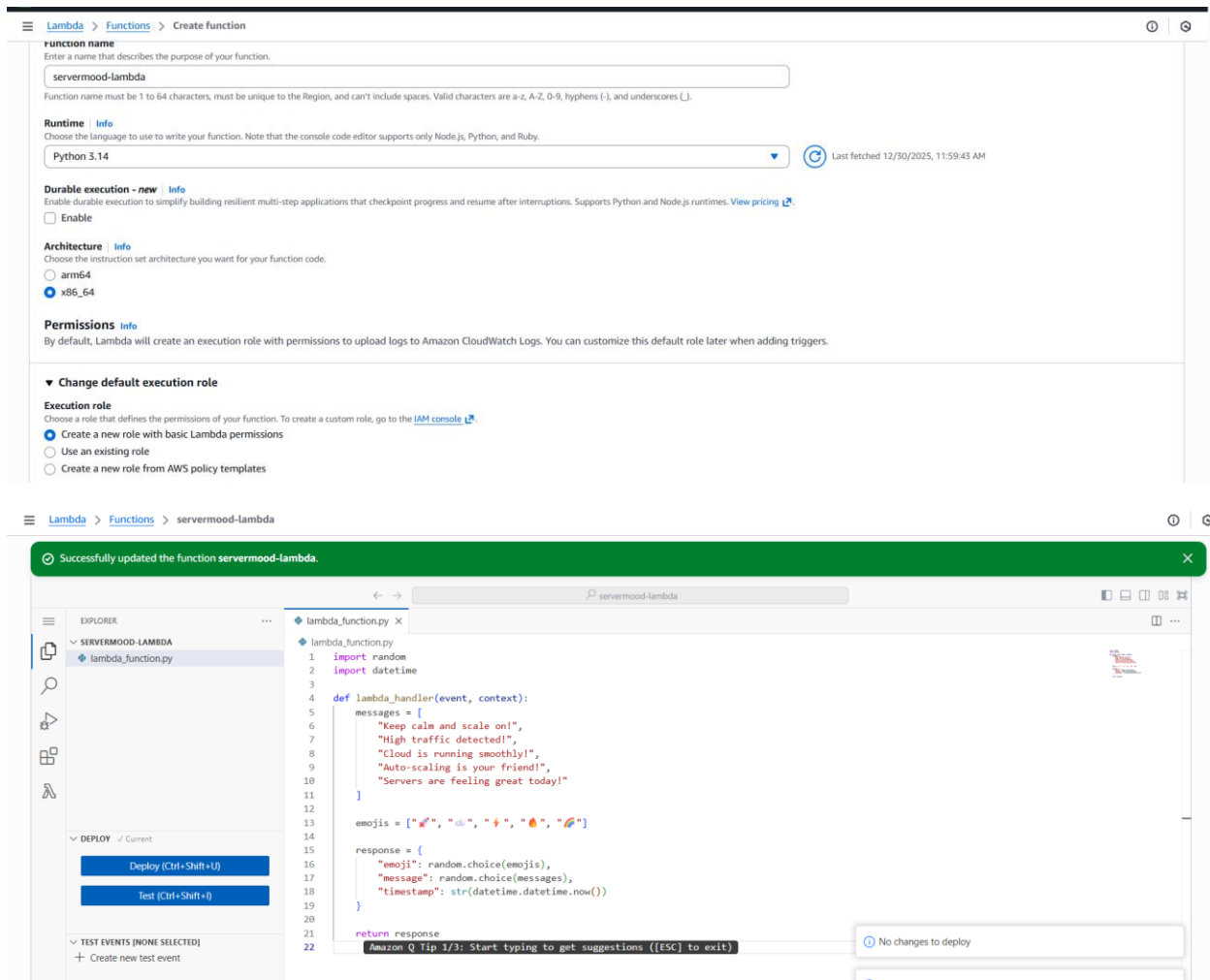
Select an existing statement in the policy or add a new statement.

[+ Add new statement](#)

4. Create the Lambda Function

- Lambda → Create function
- Name: ServerMoodLambda
- Runtime: Python 3.14

- Execution role: Create new role



5. Launch EC2 Instances(Web-Server-1)

- Security Group for EC2
- Allow HTTP (80) only from ALB Security Group
- AMI: Amazon Linux 2
- Instance type: t2.micro
- Subnet: Private-Subnet-1
- IAM role: EC2-Lambda-Invoke-Role
- No public IP
- Security group: EC2-SG

Add user data (Mainly)

Name and tags [Info](#)

Name

Web-Server-1

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Quick Start



[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-00ca570c1b6d79f36 (64-bit (x86), uefi-preferred) / ami-061d45d4bd9c71ba1 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)
ami-00ca570c1b6d79f36

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs. 750 hours per month of

[Cancel](#)

[Launch instance](#)

[Preview code](#)

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-00ca570c1b6d79f36

Publish Date

2025-12-03

Username

ec2-user

[Verified provider](#)

▼ Instance type [Info](#) [Get advice](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0268 USD per Hour On-Demand Linux base pricing: 0.0124 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour

☒ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Proceed without a key pair (Not recommended)

Default value

[Create new key pair](#)

▼ Network settings [Info](#)

[Edit](#)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-0bac0181adad11c0c (servermood-vpc)

10.0.0.0/16

Subnet [Info](#)

subnet-095943221dd2640a8

VPC: vpc-0bac0181adad11c0c

Zone type: Availability Zone

IP addresses available: 4091

CIDR: 10.0.128.0/20

servermood-subnet-private1-ap-south-1a

Owner: 989615686095

Availability Zone: ap-south-1a (aps1-az1)

[Create new subnet](#)

Auto-assign public IP [Info](#)

Disable

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups [Info](#)

Select security groups

Ec2-SG sg-045e3c68912a96e1d

VPC: vpc-0bac0181adad11c0c

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Number of instances [Info](#)

1

t2.micro

Firewall (security group)

Ec2-SG

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs. 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

[Cancel](#)

[Launch instance](#)

[Preview code](#)

▼ Advanced details [Info](#)

Domain join directory [Info](#)

Select

[Create new directory](#)

IAM instance profile [Info](#)

EC2-Lambda-Invoke-Role

arn:aws:iam::989615686095:instance-profile/EC2-Lambda-Invoke-Role

[Create new IAM profile](#)

Launch EC2 – Web-Server-2

- Security Group for EC2
- Allow HTTP (80) only from ALB Security Group
- AMI: Amazon Linux 2
- Instance type: t2.micro
- Subnet: Private-Subnet-2
- IAM role: EC2-Lambda-Invoke-Role
- No public IP
- Security group: EC2-SG

EC2 > Instances > Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name
Web-Server-2

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Instance type

Instance type
t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Windows base pricing: 0.017 USD per Hour On-Demand RHEL base pricing: 0.0268 USD per Hour On-Demand Linux base pricing: 0.0124 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour On-Demand SUSE base pricing: 0.0124 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
Proceed without a key pair (Not recommended) Default value Create new key pair

Network settings

Network
vpc-08759b5fe29f72e8d

Edit

Summary

Number of instances 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-00ca570c1b6d79f36

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs. 750 hours per month of

Cancel Launch instance Preview code

Summary

Number of instances 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.9.2...read more
ami-00ca570c1b6d79f36

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs. 750 hours per month of

Cancel Launch instance

EC2 > Instances > Launch an instance

Network settings

VPC - required [Info](#)

vpc-0bac0181adad11c0c (servermood-vpc) [↕](#)

Subnet [Info](#)

subnet-04a6389902dbbf65d servermood-subnet-private2-ap-south-1b [↕](#) [Create new subnet](#)

Auto-assign public IP [Info](#)

Disable [↕](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups

Ec2-SG sg-045e3c68912a96e1d [✕](#) [↕](#) [Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.9.2...[read more](#)

ami-00ca570c1b6d79f36

Virtual server type (instance type)

t2.micro

Firewall (security group)

Ec2-SG

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier 3Mbit 750 hours per month of

[Cancel](#) [Launch instance](#) [Preview code](#)

▼ Advanced details [Info](#)

Domain join directory [Info](#)

Select [↕](#) [Create new directory](#)

IAM instance profile [Info](#)

EC2-Lambda-Invoke-Role [↕](#) [Create new IAM profile](#)

arn:aws:iam::989615686095:instance-profile/EC2-Lambda-Invoke-Role

6. Create Application Load Balancer (ALB)

Create Target group first

- Target type: Instance
- Protocol: HTTP
- Port: 80
- Register both EC2 instances

EC2 > Target groups > Create target group

Step 1
Review and create

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.
servermood

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 10/32

Protocol

Protocol for communication between the load balancer and targets.
HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.
80
1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.
-

EC2 > Target groups > Create target group

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Register targets - recommended

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (2/2)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone	Private IP
☒	i-039ac44ec817036ba	Web-server-2	Running	Ec2-SG	ap-south-1b	10.0.147.23
☒	i-0975204d7262ee4de	Web-Server-1	Running	Ec2-SG	ap-south-1a	10.0.128.20

2 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.
80
1-65535 (separate multiple ports with comma)

Include as pending below

2 selections are now pending below. Include more or register targets when ready.

EC2 > Target groups > servermood

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

2
Total targets

0
Healthy

0
Unhealthy

2
Unused

0
Initial

0
Draining

0 Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (2)

Anomaly mitigation: Not applicable

Deregister

Register targets

Filter targets

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri...	Launch...
☐	i-039ac44ec817036ba	Web-server-2	80	ap-south-1b (a...	Unused	Target group is not co...	-	-	December...
☐	i-0975204d7262ee4de	Web-Server-1	80	ap-south-1a (a...	Unused	Target group is not co...	-	-	December...

Create Application Load balancer

- Type: Application Load Balancer
- Scheme: Internet-facing
- Subnets: Public-Subnet-1 & 2
- Security group: Allow HTTP (80) from anywhere
- Target group – *servermood*
- Listener - HTTP:80 → Forward to target group

EC2 > Load balancers > Compare and select load balancer type

①

Application Load Balancer [Info](#)

Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)

Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

[Create](#)

Gateway Load Balancer [Info](#)

Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

EC2 > Load balancers > Create Application Load Balancer

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

servermood

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

Network mapping [Info](#)

Map subnets to the load balancer to specify which subnets the load balancer can use to route traffic to your target instances.

EC2

>

Load balancers

>

Create Application Load Balancer

Network mapping

Info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC

Info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0bac0181adad11c0c (servermood-vpc)

10.0.0.0/16

Create VPC

IP pools

Info

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets

Info

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-032313415fa1e3299

IPv4 subnet CIDR: 10.0.0.0/20

servermood-subnet-public1-ap-south-1a

☒ ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0342a45fd90c6fcd8

IPv4 subnet CIDR: 10.0.16.0/20

servermood-subnet-public2-ap-south-1b

Security groups

Info

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

Security groups

Select up to 5 security groups

ALB-SG

sg-0c40f9249ee624250

VPC: vpc-0bac0181adad11c0c

EC2

>

Load balancers

>

Create Application Load Balancer

Listeners and routing

Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80

Remove

Protocol

HTTP

Port

80

1-65535

Default action

Info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group

Info

Choose a target group and specify routing weight or [create target group](#).

Target group

servermood

HTTP

Weight

1

Percent

100%

+ Add target group

You can add up to 4 more target groups.

Target group stickiness

Info

Make sure you have installed apache and CLI on both server

sudo yum install httpd php aws-cli

Php is used for creating our execution file to call lambda function.

```
[ec2-user@ip-10-0-128-168 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Tue 2025-12-30 11:13:16 UTC; 52min ago
     Docs: man:httpd.service(8).
    Main PID: 31135 (httpd)
   Status: "Total requests: 344; Idle/Busy workers 100/0;Requests/sec: 0.109; Bytes served/sec: 49 B/sec"
     Tasks: 230 (limit: 1120)
    Memory: 19.6M
       CPU: 2.629s
    CGroup: /system.slice/httpd.service
            └─31135 /usr/sbin/httpd -DFOREGROUND
              └─31136 /usr/sbin/httpd -DFOREGROUND
                └─31137 /usr/sbin/httpd -DFOREGROUND
                  └─31138 /usr/sbin/httpd -DFOREGROUND
                    └─31139 /usr/sbin/httpd -DFOREGROUND
                      └─31317 /usr/sbin/httpd -DFOREGROUND

Dec 30 11:13:16 ip-10-0-128-168.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Dec 30 11:13:16 ip-10-0-128-168.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Dec 30 11:13:16 ip-10-0-128-168.ap-south-1.compute.internal httpd[31135]: Server configured, listening on: port 80
[ec2-user@ip-10-0-128-168 ~]$
```

Test Lambada integration through CLI

```
aws lambda invoke \
--function-name server-mood-lambda \
--payload '{}' \
/tmp/test.json
```

```
[ec2-user@ip-10-0-128-168 ~]$ aws lambda invoke \
--function-name servermood-lambda \
--payload '{}' \
/tmp/test.json
{
  "StatusCode": 200,
  "ExecutedVersion": "$LATEST"
}
[ec2-user@ip-10-0-128-168 ~]$
```

Check its running

```
[ec2-user@ip-10-0-128-168 ~]$ cat /tmp/test.json
{"emoji": "☁️", "message": "Cloud is running smoothly!", "timestamp": "2025-12-30 12:08:13.637820"}[ec2-user@ip-10-0-128-168 ~]$
```

Crate an index.php file for expected output

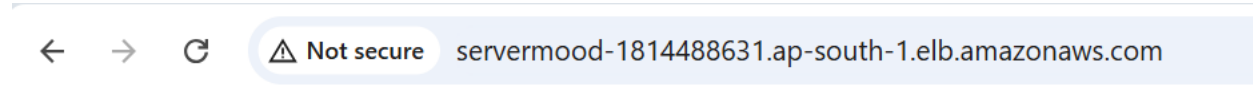
```
sudo vi /var/www/html/index.php
```

```
GNU nano 8.3 index.php Modified
<?php shell_exec("aws lambda invoke \
--function-name servermood-lambda \
--payload '{}' \
/tmp/response.json");
$response = json_decode(file_get_contents("/tmp/response.json"), true);
echo "<h2>☁️Server Mood</h2>";echo "<b>Server:</b> ".gethostname()."<br><br>";echo "<b>Lambda Message:</b><br>";echo $response['message']."<br>";echo "<b>Time:</b> ".$response['timestamp'];
```

Make sure you called Lambada function that we have created.

Then Restart Apache Server
`sudo systemctl restart httpd`

Test the Application and see results
`http:<<ALB-DNS>>`



Server Mood

Server: ip-10-0-128-168.ap-south-1.compute.internal

Lambda Message:

Cloud is running smoothly!

Time: 2025-12-30 11:58:12.570802



Server Mood

Server: ip-10-0-155-26.ap-south-1.compute.internal

Lambda Message:

Auto-scaling is your friend!

Time: 2025-12-30 12:18:13.386022

← → ↻ ⚠ Not secure servermood-1814488631.ap-south-1.elb.amazonaws.com

😄 Server Mood

Server: ip-10-0-128-168.ap-south-1.compute.internal

Lambda Message:

High traffic detected!

Time: 2025-12-30 12:18:49.163895

← → ↻ ⚠ Not secure servermood-1814488631.ap-south-1.elb.amazonaws.com

😄 Server Mood

Server: ip-10-0-128-168.ap-south-1.compute.internal

Lambda Message:

Servers are feeling great today!

Time: 2025-12-30 12:19:38.445812